

Reliable AI and Human Review

FROM EXPLANATION EVIDENCE TO CONTROLLED ACTION

Every request must be accepted, deferred for human review, or refused by a declared rule.

OBJECTIVE

The course path: selection -> explanation -> control

Session 3 inspected the chosen model. Session 4 decides when that model is allowed to act.

Session 2

A legitimate model was selected using class-balanced metrics.

Session 3

The model was opened: global reliance, local cases, and errors were inspected.

Session 4

The prediction is routed through calibration, policy, contract, anomaly, and audit evidence.

Core bridge: “Why did it predict this?” becomes “Should this prediction be acted on?”

The operational words used in Session 4

Each term is a control layer around the same model that was explained in Session 3.

Calibration

A probability check. A 0.80 confidence score should behave like about 80% correctness over similar cases.

Policy

A declared rule that says which cases may be automated and which cases must be reviewed.

Contract

A hard request schema: required fields, numeric types, and valid ranges are enforced before the model is called.

Anomaly

A valid input that is still unusual compared with historical operating patterns.

Router

A deterministic state machine that returns exactly one explicit status.

Reliability is not one metric. It is the combined behavior of all control layers.

More terms that appear in the notebook outputs

These definitions prevent the output tables from becoming unclear labels.

Evidence

The stored facts used to justify a decision: status, reason, confidence, thresholds, and version IDs.

Coverage

The share of requests allowed to receive automatic AI decisions under a chosen threshold.

Selective accuracy

Accuracy measured only on cases that were accepted for automation.

Review rate

The share deferred to humans.

Refusal

A deliberate non-prediction caused by invalid input, not a low-confidence prediction.

Audit record

A compact, reproducible record of what was checked and why the final route was selected.

Review threshold

The minimum confidence required before automatic action is allowed.

The notation used across Sessions 3 and 4

The symbols are introduced once, then reused in every equation slide.

Symbol	Meaning	Plain-language interpretation
x	one input record	one region-month request sent to the model
x_j	feature j inside x	one measured field such as rainfall or humidity
y	true class label	the known LOW / MEDIUM / HIGH class in evaluation
c	class index	one candidate output class
C	number of classes	LOW, MEDIUM, HIGH $\rightarrow C = 3$
$\hat{p}_c(x)$	predicted probability	model confidence for class c
$c^*(x)$	predicted class	the class with the largest predicted probability

The reliability symbols used in Session 4

These symbols connect calibration, policy, validation, anomaly, and routing.

Symbol	Meaning	Plain-language interpretation
T	temperature	softening/sharpening value used during calibration
$q_c(T)$	calibrated probability	probability after temperature scaling
τ	review threshold	minimum confidence for automatic acceptance
$a_i(\tau)$	accept indicator	1 if case i is automatically accepted, else 0
ℓ_j, u_j	lower and upper bounds	valid range for feature j
$s(x)$	anomaly score	how familiar the input looks to the anomaly model
θ_{anom}	anomaly threshold	score boundary below which human review is required
Q_α	quantile	the α tail cut used to define rare cases

One request will be followed through every equation

A single example makes the bridge from explanation to routing concrete.

Case

Country-month: Alexandria coastal district, July. Actual class later observed: HIGH.

Model output

$\hat{p}(\text{LOW}, \text{MEDIUM}, \text{HIGH}) = (0.14, 0.24, 0.62)$.

Initial decision

The model predicts HIGH because 0.62 is the maximum probability.



validate



explain



calibrate



threshold



contract



anomaly



route



evidence

The same prediction is not blindly trusted; it is converted into a controlled route.

Class-balanced validation before explanation

A model should not be explained until it has first been validated.

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c}$$
$$F_{1,\text{macro}} = \frac{1}{C} \sum_{c=1}^C F_{1,c}$$

For the HIGH class: P=0.78 and R=0.70, so $F_{1,\text{HIGH}}=0.738$. If class F_1 values are 0.65, 0.66, 0.738, macro $F_1 \approx 0.683$.

Symbols and use

Symbols: P_c = precision for class c. R_c = recall for class c. $F_{1,c}$ = balanced score for class c. C = number of classes. $\sum^c$ means the sum over all classes.

Why it is used: This equation was used in Session 3 to prove that LOW, MEDIUM, and HIGH were all treated seriously before explanations were discussed.

Session 4 starts from a legitimate model; it does not replace validation.

The output being explained must be the output selected

The router in Session 4 also starts from the maximum class probability.

$$c^*(x) = \arg \max_{c \in \{1, \dots, C\}} \hat{p}_c(x)$$

For $\hat{p}=(0.14, 0.24, 0.62)$, the largest value is 0.62, so $c^*(x)=\text{HIGH}$.

Symbols and use

Symbols: x = the current request. c = candidate class. C = total number of classes. $\hat{p}_c(x)$ = model probability for class c . $\arg\max$ returns the class with the largest value.

Why it is used: Session 3 used this to target the correct local explanation. Session 4 uses the same confidence value, but it is no longer treated as automatic permission to act.

The selected class is only the beginning of the route, not the final operational decision.

Global reliance: what mattered across many cases?

A globally useful feature can still be unsafe on invalid or unusual inputs.

$$I_j = \frac{1}{R} \sum_{r=1}^R [S_0 - S_{j,r}^{\text{perm}}]$$

If baseline macro F_1 $S_0=0.683$ and shuffled drought scores are 0.61, 0.63, 0.60, then $I_{\text{drought}} \approx 0.070$.

Symbols and use

Symbols: I_j = importance of feature j . R = number of shuffle repeats. S_0 = original validation score. $S^{\text{perm}}_{j,r}$ = score after feature j was shuffled in repeat r .

Why it is used: This measured model reliance. It did not prove causality, and it did not prove that every future input should be trusted.

Importance is evidence about model behavior, not permission to automate every prediction.

Local evidence: what moved one prediction?

A local explanation can be kept in the audit record for accepted cases.

$$\Delta_{j,c^*} = \hat{p}_{c^*}(x) - \hat{p}_{c^*}(x \text{ with } x_j = b_j)$$

For HIGH: original $\hat{p}=0.62$. Replacing drought_index with a typical value gives 0.45. $\Delta=0.17$, so the original drought value supported HIGH.

Symbols and use

Symbols: $\Delta_{\{j,c^*\}}$ = local effect of feature j for the predicted class. $\hat{p}_{\{c^*\}}(x)$ = original predicted-class probability. b_j = training reference value for feature j .

Why it is used: This explained a single case. Session 4 can store it as evidence, but the route is still decided by contract, anomaly, and confidence policy.

Explanation supports review. It does not remove the need for routing controls.

Formal credit allocation still stays inside the model

SHAP explains model output movement; it does not prove real-world causality.

$$\phi_{j,c} = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(M - |S| - 1)!}{M!} [v_c(S \cup \{j\}) - v_c(S)]$$

Feature j receives average marginal contribution credit across possible feature coalitions.

Symbols and use

Symbols: $\phi_{j,c}$ = SHAP value for feature j and class c. F = all features. M = number of features. S = a subset of features not containing j. $v_c(S)$ = class-c model output when only S is known.

Why it is used: This was used to assign class-specific contribution credit without claiming that a feature caused the real event.

Session 4 boundary

Even if an accepted case has an explanation, the router state is decided by validation, anomaly, and confidence policy.

Baseline plus contributions reconstructs model output

The explanation remains inside the model output space.

$$f_c(x) = \mathbb{E}[f_c(X)] + \sum_{j=1}^M \phi_{j,c}$$

Example: baseline score 0.10 + drought 0.42 + heat 0.22 - humidity 0.09 + others 0.15 = class score 0.80.

Symbols and use

Symbols: $f_c(x)$ = class-c model output. $\mathbb{E}[f_c(X)]$ = average background output. $\sum_j$ = sum over all features. $\phi_{j,c}$ = contribution from feature j to class c .

Why it is used: This equation explains how the selected score was reconstructed. It does not decide whether the prediction should be accepted automatically.

Proper wording: a feature moved the class score upward or downward.

Stability check: would top factors survive another background?

A fragile explanation should not be overclaimed.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

If $A=\{\text{drought, heat, water, humidity, vegetation}\}$ and $B=\{\text{drought, heat, water, humidity, disease}\}$, then $J=4/6 \approx 0.667$.

Symbols and use

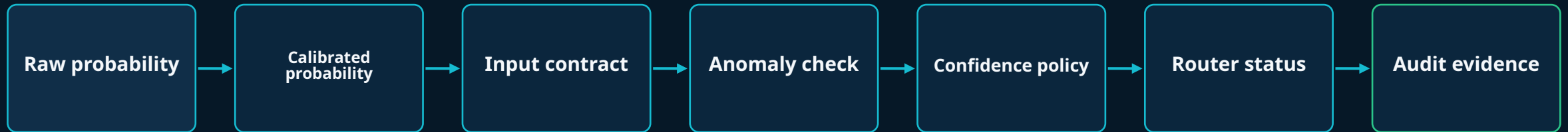
Symbols: A and B = two top-feature sets. n = shared features. u = all unique features. $|\cdot|$ = number of elements. $J(A,B)$ = overlap ratio.

Why it is used: This tests whether explanation rankings are stable enough to discuss. Session 4 still requires operational controls before action.

Stable enough to discuss is not the same as safe enough to automate.

Explanation answers “why”; reliability asks “should it act?”

Session 4 wraps the explained model with explicit operational controls.



The model is not deleted. It is governed.

Confidence must behave like evidence, not decoration

The largest probability is a score; calibration checks whether the score is trustworthy.

Raw confidence

The model says HIGH with 0.62 confidence. This number may be overconfident or underconfident.

Calibration question

When the model says about 0.60 confidence, is it correct about 60% of the time?

Operational result

The probability can be used in a policy only after its behavior has been measured.

Calibration does not guarantee improvement; it is measured evidence.

Temperature scaling adjusts probability sharpness

It is fitted on calibration data, then tested on unseen data.

$$q_c(T) = \frac{\exp(z_c/T)}{\sum_k \exp(z_k/T)}$$
$$z_c = \log(p_c + \epsilon)$$

If $T > 1$, probabilities usually become softer. For $p(\text{HIGH}) = 0.62$, calibrated $q(\text{HIGH})$ may become 0.56.

Symbols and use

Symbols: $q_c(T)$ = calibrated probability for class c . z_c = log-transformed raw probability. T = temperature. ϵ = tiny constant to avoid $\log(0)$. k = all classes in the denominator.

Why it is used: This equation changes probability sharpness without changing the underlying class order in most cases.

Temperature is a calibration control, not a new disease-risk model.

The temperature is selected by minimizing log loss

The calibration year is used for fitting; the test year remains untouched.

$$T^* = \arg \min_T \left[-\frac{1}{N} \sum_{n=1}^N \log q_{y_n}^{(n)}(T) \right]$$

If $T=1.30$ gives lower calibration log loss than $T=1.00$, then $T^*=1.30$ is selected.

Symbols and use

Symbols: T^* = selected temperature. N = number of calibration records. y_n = true label of record n . $q_{\{y_n\}^{(n)}}(T)$ = calibrated probability assigned to the correct class.

Why it is used: This chooses the temperature that gives better probability behavior on calibration data.

The test set is used to evaluate the decision engine, not to choose T .

Reliability metrics measure probability behavior

ECE and Brier answer different reliability questions.

$$\text{ECE} = \sum_b \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$$

If a bin has average confidence 0.56 and accuracy 0.50, its gap is 0.06. Weighted bin gaps are summed.

$$\text{Brier} = \frac{1}{N} \sum_{n=1}^N \sum_c \left(q_c^{(n)} - \mathbf{1}[y_n = c] \right)^2$$

If the true class is HIGH, the target vector is (0,0,1); squared probability errors are summed.

Symbols and use

Symbols: B_b = records inside confidence bin b . $\text{acc}(B_b)$ = observed accuracy in that bin. $\text{conf}(B_b)$ = mean confidence in that bin. $q_c^{(n)}$ = probability assigned to class c for record n . $\mathbf{1}[y_n=c] = 1$ when c is the true label and 0 otherwise.

Why it is used: ECE checks calibration gaps by bins. Brier checks squared probability error across all classes.

The same case after calibration

The prediction is still HIGH, but confidence is now treated more carefully.

Stage	LOW	MEDIUM	HIGH
Raw $\hat{p}$	0.14	0.24	0.62
Calibrated $q(T^*)$	0.18	0.26	0.56
Selected class	-	-	HIGH

ECE reading
Lower ECE means confidence bins are closer to observed correctness.

Brier reading
Lower Brier means probabilities are closer to the true one-hot label.

Policy impact
The confidence used by the router is $q(\text{HIGH})=0.56$.

A review policy turns probability into an action rule

A threshold should be declared before the final test outcome is interpreted.

Policy requirement

Example: at least 25% coverage and at least 82% selective accuracy on calibration data.

Review threshold

τ is the minimum confidence required for automatic acceptance.

Human capacity

If reviewers can handle 30% of requests, coverage should not drop below 70%.

Policy makes the trade-off explicit instead of hiding it inside model accuracy.

Coverage says how much automation remains

Higher thresholds usually improve accepted-case quality but reduce automation.

$$a_i(\tau) = \mathbf{1}[\text{confidence}_i \geq \tau]$$
$$\text{coverage}(\tau) = \frac{1}{N} \sum_i a_i(\tau)$$

If 70 out of 100 records exceed τ , then coverage=0.70 and review rate=0.30.

Symbols and use

Symbols: $a_i(\tau)$ = accept indicator for case i . confidence_i = calibrated maximum probability. τ = review threshold. N = number of cases. $\sum_i$ sums over all cases.

Why it is used: This equation measures the share of cases allowed to bypass human review.

Coverage is an operations metric, not a model-performance metric.

Selective accuracy measures accepted cases only

The model is judged on the cases it was allowed to decide automatically.

$$\text{selective accuracy}(\tau) = \frac{\sum_i a_i(\tau) \mathbf{1}[\hat{y}_i = y_i]}{\sum_i a_i(\tau)}$$

If 70 cases are accepted and 60 are correct, selective accuracy=60/70=85.7%.

Symbols and use

Symbols: $\hat{y}_i$ = predicted class for case i . y_i = true class. $\mathbf{1}[\hat{y}_i=y_i]$ = 1 if prediction is correct. $a_i(\tau)$ = accepted-case flag.

Why it is used: This metric tells whether the automatic portion is trustworthy enough under the declared policy.

A router can be useful even when not every case is automated.

Threshold trade-off for the example policy

The selected threshold must satisfy both accuracy and review-capacity needs.

τ	Coverage	Review rate	Selective accuracy	Policy decision
0.55	80%	20%	81.0%	fails accuracy
0.60	72%	28%	84.0%	passes
0.65	63%	37%	87.0%	too much review
0.70	51%	49%	90.0%	too much review

Chosen policy

$\tau = 0.60$ balances quality with review capacity.

Our case

calibrated confidence = 0.56, so it falls below τ .

The case is not refused. It is deferred because confidence is below the declared policy.

Invalid measurements are rejected before prediction

A contract prevents impossible requests from becoming dramatic model outputs.

$$\text{valid}(x) = \bigwedge_j [x_j \in \mathbb{R} \wedge \ell_j \leq x_j \leq u_j]$$

Temperature must be numeric and within its allowed range; rainfall must not be negative; month must be 1 to 12.

Symbols and use

Symbols: $\text{valid}(x)$ = whether the full request passes validation. $\wedge_j$ = all features must pass. ℓ_j = lower bound. u_j = upper bound. x_j = measured feature value.

Why it is used: This equation enforces the external request schema before feature engineering or model prediction.

Reject means no prediction is produced.

The contract separates impossible from merely uncertain

A rejected input should not receive a class prediction.

Field	Allowed range	Valid case	Attack case
temperature_c	-15 to 60	34	150
rainfall_mm	0 to 500	18	-4
humidity_pct	0 to 100	72	130
month	1 to 12	7	13

Valid case

The model may be called after the contract passes.

Attack case

The route is REJECT_INVALID_INPUT and prediction=None.

Valid can still be historically unusual

The anomaly layer guards the boundary between valid syntax and familiar operating conditions.

$$\theta_{\text{anom}} = Q_{\alpha}(\{s(x_i)\})$$
$$\text{review}_{\text{anom}}(x) = \mathbf{1}[s(x) < \theta_{\text{anom}}]$$

If the 2% calibration quantile is $\theta = -0.03$ and the current score is -0.12 , the case is reviewed as anomalous.

Symbols and use

Symbols: $s(x)$ = anomaly score from the anomaly model. Q_{α} = α quantile. θ_{anom} = anomaly threshold. $\mathbf{1}[\cdot]$ = indicator that returns 1 when condition is true.

Why it is used: The lower score tail defines cases that are valid but too unfamiliar for automatic action.

Anomaly review is not a punishment; it is a safety boundary.

Two valid cases can receive different routes

The unusual valid case passes the contract but fails familiarity.

Case	Contract	Anomaly score $s(x)$	Threshold θ	Anomaly result
Normal July	valid	0.12	-0.03	not anomalous
Strange dry-humid combo	valid	-0.12	-0.03	human review
Impossible attack	invalid	not computed	-	rejected earlier

Important order

Invalid cases are rejected before anomaly scoring.

Review reason

Anomaly review means the input is legal but unfamiliar.

Exactly one explicit state must be returned

The order is deterministic: invalid first, then anomaly, then low confidence, then accept.

```
route(x) =  
  REJECT, if not valid(x)  
  REVIEW_ANOMALY, if  $s(x) < \theta_{\text{anom}}$   
  REVIEW_LOW_CONFIDENCE, if  $\text{confidence}(x) < \tau$   
  ACCEPT, otherwise
```

This is a state machine, not a suggestion list. The first true condition decides the route.

Symbols and use

Symbols: $\text{route}(x)$ = final status. $\text{valid}(x)$ = input contract result. $s(x)$ = anomaly score. θ_{anom} = anomaly threshold. $\text{confidence}(x)$ = calibrated maximum class probability. τ = review threshold.

Why it is used: The router converts model output into operational behavior that can be tested and audited.

The Alexandria case reaches one final state

All equations now meet in one routed decision.

Layer	Value	Decision
Prediction target	$c^*(x)=HIGH$	candidate class selected
Explanation	drought effect $\Delta=0.17$	reason can be recorded
Calibration	$q(HIGH)=0.56$	confidence softened
Contract	$valid(x)=true$	model request allowed
Anomaly	$s(x)=0.12 > -0.03$	not anomalous
Policy	$0.56 < \tau=0.60$	not accepted automatically

Final route

HUMAN_REVIEW_LOW_CONFIDENCE

Interpretation

The model may be right, but automatic action is not allowed under the policy.

Four test cases prove the state machine

A reliable decision engine should demonstrate every intended status.

Accept

valid, non-anomalous,
confidence $0.74 \geq \tau$
ACCEPT_AI_DECISION

Low confidence

valid, non-anomalous,
confidence $0.56 < \tau$
HUMAN_REVIEW_LOW_CONFIDENCE

Anomaly

valid, but $s(x) < \theta$
HUMAN_REVIEW_ANOMALY

Reject

invalid fields before model
call
REJECT_INVALID_INPUT

The notebook should not end until all four states are shown.

The final output must explain the route, not only the class

Evidence makes the decision reproducible, reviewable, and testable.

```
{
  request_id: "ALEX-2025-07",
  prediction: "HIGH",
  confidence: 0.56,
  status: "HUMAN_REVIEW_LOW_CONFIDENCE",
  reason: "confidence_below_policy",
  review_threshold: 0.60,
  anomaly_score: 0.12,
  anomaly_threshold: -0.03,
  policy_version: "reliability_policy_v1"
}
```

What evidence proves

The state was selected by a declared rule, not by improvisation.

What evidence avoids

A vague answer such as "the AI was unsure" without thresholds or versions.

How each output should be explained

The notebook should be narrated as a decision-engine build, not as disconnected cells.

Output	What it proves	What not to claim
Calibration metrics	probability behavior was measured	that calibration always improves
Threshold table	automation-review trade-off was declared	that one threshold is universal
Input contract test	invalid requests are blocked before prediction	that rejected cases are low confidence
Anomaly score	valid-but-unusual cases are deferred	that anomaly means wrong
Router record	one explicit state is returned	that the model alone made the final decision

A 150-minute section that stays coherent

The equations should be introduced only when they are needed for the decision engine.

Time	Activity	Main evidence
0-15	Bridge from Session 3 and symbol map	why explanation is not certification
15-45	Calibration and reliability metrics	ECE, Brier, temperature
45-75	Threshold policy	coverage and selective accuracy
75-105	Contract and anomaly review	reject vs review
105-135	Router implementation	four explicit states
135-150	Evidence record and assignment	auditable decision output

Before delivery, the section should pass these checks

The presentation and notebook are considered ready only when each point is true.

Equation QA

Every equation has symbols, purpose, and a numerical interpretation.

Route QA

ACCEPT, LOW_CONFIDENCE, ANOMALY, and REJECT are all demonstrated.

Output QA

Each major notebook output has a sentence explaining what it proves.

Visual QA

No clipped text, no hidden checklist items, and no unreadably small code.

Module QA

src modules are discovered from the uploaded package before execution.

Evidence QA

The final artifact includes thresholds, reasons, and version identifiers.

Build a reliable AI decision router

The output is a decision engine with evidence, not a naked classifier.

1. Calibrate

Report temperature, ECE, and Brier before and after calibration.

2. Choose policy

Build the threshold table and declare the selected threshold.

3. Enforce contract

Reject impossible inputs before prediction.

4. Add review

Defer anomalous or low-confidence cases to human review.

5. Record evidence

Return exactly one status with reason, thresholds, and versions.

Session result: the classifier now knows when to accept, defer, and refuse.

Session 3 symbols, placed directly below each equation

These equations inspected the selected model before Session 4 allowed any action.

F1 and Macro F1

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c}$$

$$F_{1,\text{macro}} = \frac{1}{C} \sum_{c=1}^C F_{1,c}$$

Symbols: P_c precision; R_c recall; $F_{1,c}$ class score; C number of classes; $\sum$ sum over classes.
Use: The model was checked class by class before explanation.

Prediction target

$$c^*(x) = \arg \max_{c \in \{1, \dots, C\}} \hat{p}_c(x)$$

Symbols: x request; c class; C total classes; $\hat{p}_c(x)$ predicted probability; argmax selects the largest value.
Use: The explained class is the class actually selected by the model.

Jaccard stability

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Symbols: A, B top-factor sets; n shared items; u all unique items; $|\cdot|$ count; J overlap ratio.
Use: A local explanation was checked for stability before being overclaimed.

Session 4 continuation

The model can be explained, but explanation alone is not automation permission.

Global and local explanation equations

These equations produce evidence that can be stored, not automatic permission to act.

Permutation importance

$$I_j = \frac{1}{R} \sum_{r=1}^R [S_0 - S_{j,r}^{\text{perm}}]$$

Symbols: I_j importance; R repeats; S_0 baseline score; $S_{j,r}^{\text{perm}}$ score after shuffling feature j .

Use: A feature was globally relied on when shuffling it reduced validation score.

Local occlusion

$$\Delta_{j,c^*} = \hat{p}_{c^*}(x) - \hat{p}_{c^*}(x \text{ with } x_j = b_j)$$

Symbols: Δ local effect; j feature; c^* predicted class; b_j training reference; $\hat{p}$ predicted probability.

Use: One feature value was compared with a typical training reference.

Additive explanation

$$f_c(x) = \mathbb{E}[f_c(X)] + \sum_{j=1}^M \phi_{j,c}$$

Symbols: f_c output; $\mathbb{E}$ background average; M feature count; ϕ contribution; Σ sum over features.

Use: Class output was reconstructed from baseline and contributions.

Formal credit and probability calibration

Session 3 assigned credit. Session 4 measures whether confidence behaves reliably.

Shapley value

$$\phi_{j,c} = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(M - |S| - 1)!}{M!} [v_c(S \cup \{j\}) - v_c(S)]$$

Symbols: ϕ contribution; F all features; S subset before j joins; M feature count; v_c model output.

Use: Credit was averaged across possible feature coalitions.

Temperature scaling

$$q_c(T) = \frac{\exp(z_c/T)}{\sum_k \exp(z_k/T)}$$

$$z_c = \log(p_c + \varepsilon)$$

Symbols: q_c calibrated probability; T temperature; z_c log probability; ε tiny stabilizer; k classes in denominator.

Use: Raw probabilities were softened or sharpened before policy use.

Temperature objective

$$T^* = \arg \min_T \left[-\frac{1}{N} \sum_{n=1}^N \log q_{y_n}^{(n)}(T) \right]$$

Symbols: T^* selected temperature; N calibration records; y_n true class; $q_{y_n}^{(n)}$ probability assigned to the correct class.

Use: The temperature was selected on calibration data only.

Reliability metrics and threshold policy

These equations turn confidence into an automation-review trade-off.

ECE

$$\text{ECE} = \sum_b \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$$

Symbols: B_b bin; N total cases; acc bin accuracy; conf bin mean confidence.
Use: Calibration gaps were measured across confidence bins.

Brier score

$$\text{Brier} = \frac{1}{N} \sum_{n=1}^N \sum_c \left(q_c^{(n)} - \mathbf{1}[y_n = c] \right)^2$$

Symbols: q probability; n record; c class; y_n truth; indicator equals 1 for the true class.
Use: Squared probability error was measured across all classes.

Coverage

$$a_i(\tau) = \mathbf{1}[\text{confidence}_i \geq \tau]$$

$$\text{coverage}(\tau) = \frac{1}{N} \sum_i a_i(\tau)$$

Symbols: a_i accept flag; τ review threshold; confidence_i calibrated maximum probability; N cases.
Use: The automatic decision share was computed.

Selective accuracy

$$\frac{\sum_i a_i(\tau) \mathbf{1}[\hat{y}_i = y_i]}{\sum_i a_i(\tau)}$$

Symbols: $\hat{y}$ prediction; y truth; a_i accepted flag; τ threshold.
Use: Accuracy was measured only on accepted cases.

Contract, anomaly, and router equations

These equations decide whether the model may act, defer, or refuse.

Input contract

$$\text{valid}(x) = \bigwedge_j [x_j \in \mathbb{R} \wedge \ell_j \leq x_j \leq u_j]$$

Symbols: x request; x_j feature value; ℓ_j lower bound; u_j upper bound; $\wedge$ all checks must pass.

Use: Impossible inputs were rejected before prediction.

Anomaly review

$$\theta_{\text{anom}} = Q_{\alpha}(\{s(x_i)\})$$

$$\text{review}_{\text{anom}}(x) = \mathbf{1}[s(x) < \theta_{\text{anom}}]$$

Symbols: $s(x)$ anomaly score; Q_{α} alpha quantile; θ_{anom} threshold; indicator returns 1 when review is needed.

Use: Valid but unfamiliar inputs were deferred.

Router

route(x) =
 reject if not valid(x)
 review if $s(x) < \theta$
 review if $\text{conf}(x) < \tau$
 accept otherwise

Symbols: route final state; θ anomaly threshold; τ review threshold.

Final rule: explanation supports evidence; routing controls action.